

Watchy v2 — Test Point Identification

Multimeter procedure for the GDEY0154D67-FL04 frontlight mod, I²C driver variant. Applies to the official v2.0 board and, with the caveat in section 7, to USB-C clones built from the same schematic.

1. What you end up with

Four pads on the Watchy board, positively identified and marked: **TP4 (3V3)**, **TP5 (GND)**, **TP2 (SCL)** and **TP3 (SDA)**. Those are the only connections the frontlight adapter needs.

The v2 differs from the v3 here. There is no free, switchable GPIO on a test pad, so the frontlight is driven by a PCA9633 on the I²C bus rather than a MOSFET on a GPIO. That means four wires instead of three, but all four come from one 2×2 block of pads.

Convenient consequence: you do not need to tell SCL from SDA. Once GND and 3V3 are known, the remaining two are the I²C pair, and swapping them is harmless — the scanner simply will not find the driver at 0x62, and you swap the wires.

2. Before you start

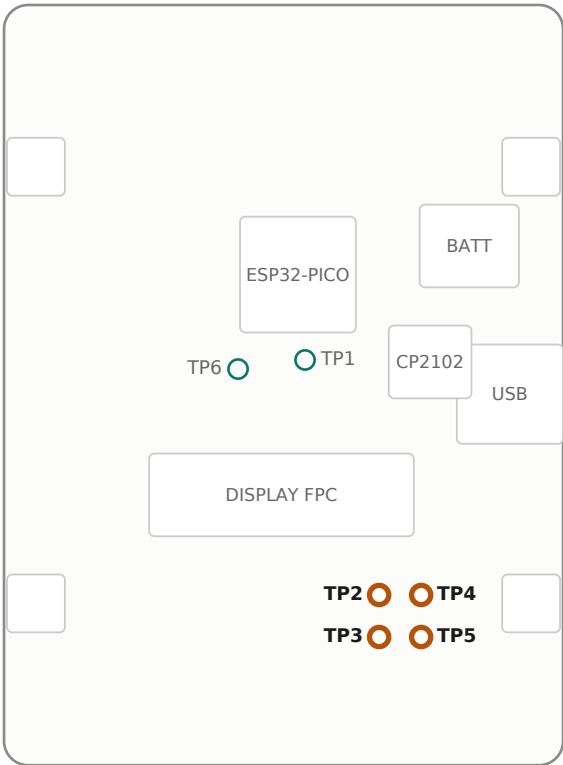
Step	Why
Disconnect the battery	Removes any short-circuit risk while probing. Pull on the connector housing, never the wires.
Black lead in COM , red in INPUT	The 10 A and μ A/mA jacks are not used anywhere in this procedure.
Clip the black lead to the USB shell	Frees one hand. The 1 mm pads need a steady probe.

Power comes from USB in phase 2 only. Phases 1 and 3 need the board de-energised.

3. Where the pads are

Component side
Board 33.8 × 46.0 mm
Pads 1 × 1 mm

**Bold outline =
needed for the mod**



Test point positions extracted from *Watchy.kicad_pcb* rev 2.0; component outlines indicative. Origin is the top-left corner of the board outline.

TP	Net	X (mm)	Y (mm)	Needed
TP1	GPIO0	18.20	21.46	—
TP2	SCL	22.68	35.67	yes
TP3	SDA	22.68	38.21	yes
TP4	+3V3	25.22	35.67	yes
TP5	GND	25.22	38.21	yes
TP6	GND	14.15	22.01	spare ground

TP2 to TP5 form a **2×2 square at 2.54 mm pitch**, near the bottom of the board below the display connector. Seen from the component side the layout is:

SCL	3V3
SDA	GND

TP1 and TP6 sit separately, between the ESP32 and the display connector. Neither is needed, and clones frequently omit them.

4. Meter settings

Phase	Rotary position	Display should show
1	Ω , then Select until the speaker icon appears	continuity beep
2	V with solid and dashed line (DC volts)	volts, autoranging
3	Ω , Select back to plain resistance	ohms, OL when open

If unsure which Select mode you are in: touch the two probes together. A beep means continuity mode; a reading near zero ohms means resistance mode.

5. Procedure

Phase 1 — find GND (board unpowered, USB out)

Continuity mode. Black probe on the USB shell, red probe on each pad in turn. **A beep means GND.** On the official v2 two test points do this, TP5 and TP6, plus the negative battery terminal.

Use the GND pad that sits inside the 2×2 block — that is TP5, and it keeps all four wires together in one place.

Phase 2 — voltages (USB in, battery still out)

DC volts. Black on GND, red on each remaining pad.

Reading	Meaning
0.00 V	another GND
3.2 – 3.4 V, rock steady	+3V3 → this is TP4
4.0 – 4.3 V	battery pad — do not solder here
around 3.3 V, less steady	a signal pad — continue with phase 3

With the battery removed the charger regulates into an open circuit, so battery pads sit near 4.2 V. That makes them easier to tell from 3V3 than they would be with a battery fitted. Confusing the two is the one mistake that destroys the frontlight LEDs: 4.2 V across three parallel white LED dies rated 2.9 V is well over 100 mA.

Phase 3 — find the I²C pair (USB out again)

Resistance mode, board de-energised — a live board gives meaningless resistance readings. One probe on the TP4 pad found in phase 2, the other on each remaining signal pad.

Reading	Meaning
around 4.7 kΩ	SCL or SDA — these are TP2 and TP3
tens of kΩ or higher	GPIO0 (TP1) — not used, ignore it
OL / open	not a signal pad

The 4.7 kΩ figure is not a guess: R2 and R3 in the v2 schematic are the I²C pull-ups to +3V3, both 4.7 kΩ. No other net on the board sits at that value, which is what makes this a positive identification rather than an elimination.

Phase 4 — confirm after wiring

Once the adapter board is connected, run an I²C scanner. You should see three devices:

Address	Device
0x18 or 0x19	BMA423 accelerometer
0x51	PCF8563 real-time clock
0x62	PCA9633 frontlight driver

If the first two appear but 0x62 does not, swap the SCL and SDA wires and scan again. No harm results from having had them the wrong way round.

6. Record the result

Signal	Test point	Position on the board	Verified
3V3	TP4		
GND	TP5		
SCL	TP2		
SDA	TP3		

Marking the four pads on a photo of your own board pays for itself when you come back to solder.

7. Clones: read this before trusting the map

SQFMI published the v2 schematic but never the PCB layout. Any USB-C Watchy from a marketplace is therefore a redraw: same circuit, different board. **Pad positions in section 3 do not transfer.**

What does transfer is the procedure. The discriminators used in phases 1 to 3 — continuity to the USB shell, a steady 3.3 V, and a 4.7 kΩ pull-up — follow from the circuit, not the layout. Work through the phases on whatever pads your board actually has.

Two further things worth confirming on a clone:

Question	How to answer it
Does the clone follow the v2 pinmap?	Flash the Watchy library with <i>ARDUINO_WATCHY_V20</i> . If display, all four buttons, vibration, accelerometer and RTC all work, the mapping matches.
Which ESP32 is fitted?	<i>esptool.py chip_id</i> , or <i>ESP.getChipModel()</i> in a sketch.

8. If something does not add up

Symptom	Likely cause
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Nothing beeps against the USB shell	Shell may be anodised or coated. Use the negative battery terminal as the ground reference instead.
No pad reads 4.7 kΩ to 3V3	Board still powered, or the clone omitted the I ² C test points. Retry de-energised, then look for pull-up resistors near the accelerometer and probe their pads instead.
Three or more pads read a steady 3.3 V	Some may be unpopulated component positions rather than test points. Phase 3 will separate them.
Resistance readings drift	The board is still powered. Unplug USB for phases 1 and 3.

Companion documents: [watchy-v2-frontlight-appendix.md](#) (circuit, BOM and firmware for this variant) and [watchy-v3-frontlight-mod.md](#) (the main build guide).